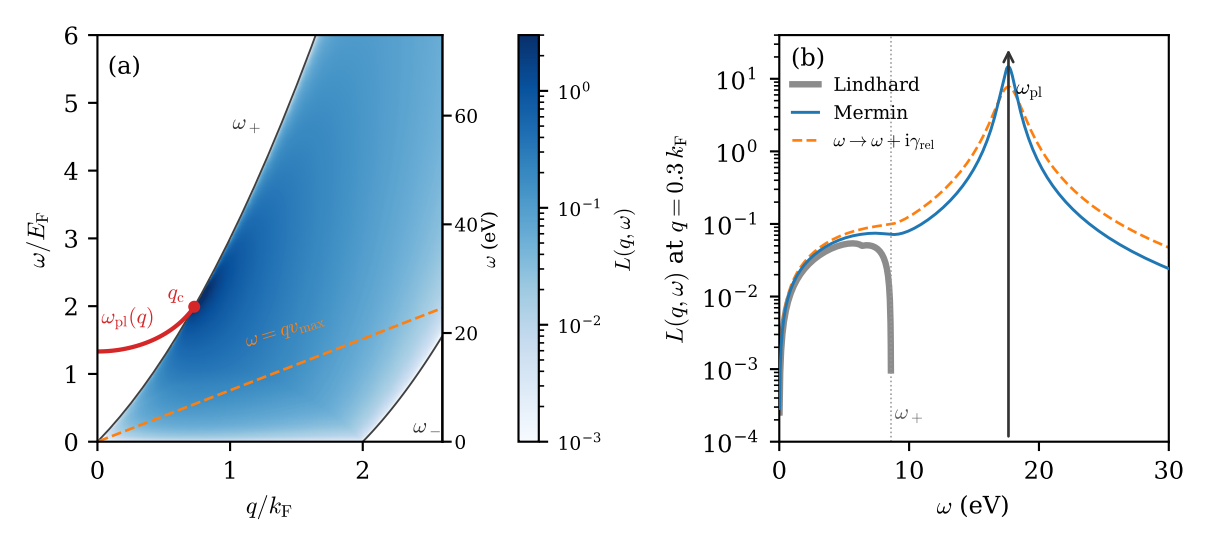

(a)
ω_+
ω_-
q_c
$\omega_{pl}(q)$
$\omega = qv_{max}$
ω/E_F
q/k_F
ω (eV)
$L(q,\omega)$
(b)
Lindhard
Mermin
$\omega \to \omega + i\gamma_{rel}$
ω_{pl}
ω_+
$L(q,\omega)$ at $q = 0.3\,k_F$
ω (eV)